Mathematical Model & `Reactome` quirks

Ionuț Cicio

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1 Kinetic constants estimation

Notation preliminaries:

- $\mathbb{N}_1 = \mathbb{N} - \{0\}$
- $\mathbb{R}^+ = \{x \mid x \in \mathbb{R} \wedge x \geq 0\}$

1.1 Model definition

Definition 1 (Problem)**.** Let $\mathrm{P} = (\mathrm{X}_{\texttt{interest}}, \mathrm{X}_{\texttt{excluded}}, D)$ be a problem where:

- $\mathrm{X}_{\texttt{interest}}$ is the set of physical entities of interest, from which the biochemical network is generated
- $\mathrm{X}_{\texttt{excluded}}$ is the set of physical entities whose input reactions can be excluded in the generation of the biochemical network
- $D = (\mathrm{X}_{\texttt{data}}, v_{\texttt{data}})$ where:
  - $\mathrm{X}_{\texttt{data}}$ is the set of physical entities for which a target value is known
  - $v_{\texttt{data}} : \mathrm{X}_{\texttt{data}} \to [0, 1]$ is the target value at equilibrium for the physical entity

Definition 2 (Biochemical network)**.** A biochemical network G is a tuple $(\mathrm{X}, \mathrm{R}, \mathrm{C}, \mathrm{E_R}, \mathrm{E_C})$ where

- X is the set of physical entities in the biochemical network; to be consisten with `Reactome` a physical entity models the instance of a chemical entity in a compartment
- R is the set of reactions in the biochemical network
  - $\mathrm{R}_{\texttt{reversible}} \subseteq \mathrm{R}$ is the subset of reactions which are reversible
- $\mathrm{E_R} : \mathrm{R} \to (\mathcal{P}(\mathrm{X} \times \mathrm{T_S}) \to \mathbb{N}_1) \cup \mathcal{P}(\mathrm{X} \times \mathrm{T_M})$ describes the roles the physical entities have in the reactions, with $\mathrm{T} = \mathrm{T_S} \cup \mathrm{T_M}$ where
  - $\mathrm{T_S} = \{\texttt{reactant}, \texttt{product}\}$
  - $\mathrm{T_M} = \{\texttt{enzyme}, \texttt{activator}, \texttt{inhibitor}\}$
- C is the set of compartments in which the species are located
- $\mathrm{E_C} : \mathrm{X} \to \mathcal{P}(\mathrm{C}) - \{\varnothing\}$ relates physical entities with their compartments; each entity must have at least one compartments

Definition 3 (Problem closure)**.** Given a problem P and the `Reactome` biochemical network G let $G' = (\mathrm{X}', \mathrm{R}', \mathrm{C}', \mathrm{E_R}', \mathrm{E_C}')$ be the closure of P in the network G where:

- $x \in \mathrm{X}_{\texttt{interest}} \Rightarrow x \in \mathrm{X}'$
- $(x \in \mathrm{X}' - \mathrm{X}_{\texttt{excluded}} \wedge \exists\, r \in \mathrm{R}, \nu \in \mathbb{N}_1\ (x, \texttt{product}, \nu) \in \mathrm{E_R}(r)) \Rightarrow r \in \mathrm{R}'$
- $(r \in \mathrm{R}' \wedge \exists\, x, \nu (x, \texttt{reactant}, \nu) \in \mathrm{E_R}(r)) \Rightarrow x \in \mathrm{X}'$
- $r \in \mathrm{R}' \Rightarrow \mathrm{E_R}'(r) = \mathrm{E_R}(r)$

1.2 Reactions rate laws

In order to describe the kinetics of a biochemical network, the rate law for each reaction must be defined.

Given a biochemical network $G = (\mathrm{X}, \mathrm{R}, \mathrm{C}, \mathrm{E_R}, \mathrm{E_C})$, let $[x]$ denote the concentration of the physical entity $x \in \mathrm{X}$, and v_r denote the rate of reaction $r \in R$.

A reaction has inputs, outputs, enzymes, activators and inhibitors. In order to handle activators and inhibitors of the reaction r let $f^r_{\text{regulation}}$ be defined as:

$$f^r_{\texttt{regulation}} = \prod_{\substack{(x,\ \texttt{activator}) \\ \in \\ \mathrm{E_R}(r)}} \frac{[x]}{K_A^{x,r} + [x]} \prod_{\substack{(x,\ \texttt{inhibitor}) \\ \in \\ \mathrm{E_R}(r)}} \frac{K_I^{x,r}}{K_I^{x,r} + [x]} \tag{1}$$

Given a reaction $r \in R$, its rate v_r can be defined based on the reversibility of the reaction and the requirement of enzymes for the reaction.

1. $r \notin R_{\texttt{reversible}} \land \neg\ \exists\, e\ (e, r) \in E_R^{\texttt{enzyme}}$

$$v_r = f^r_{\texttt{regulation}} \cdot k^r_+ \prod_{\substack{(p,r)\\ \in \\ E_R^{\texttt{reactant}}}} [p]^{\nu_{\texttt{reactant}}(p,r)} \tag{2}$$

2. $r \in R_{\texttt{reversible}} \land \neg\ \exists\, e\ (e, r) \in E_{R,\texttt{enzyme}}$

$$v_r = f^r_{\texttt{regulation}} \cdot \left(k^r_+ \prod_{\substack{(p,r)\\ \in \\ E_R^{\texttt{reactant}}}} [p]^{\nu_{\texttt{reactant}}(p,r)} - k^r_- \prod_{\substack{(p,r)\\ \in \\ E_R^{\texttt{product}}}} [p]^{\nu_{\texttt{product}}(p,r)} \right) \tag{3}$$

In Equation 2 and Equation 3 I'm not sure if $f^r_{\texttt{regulation}}$ can be written there. In literature regulators act on enzymes, but there are some reactions in Reactome with regulators and without enzymes (@).

3. $r \notin R_{\texttt{reversible}} \land \exists\, e\ (e, r) \in E_R^{\texttt{enzyme}}$

$$v_r = [e] \cdot f^r_{\texttt{regulation}} \cdot k^r_{\text{cat},+} \prod_{\substack{(p,r)\\ \in \\ E_R^{\texttt{reactant}}}} \left(\sum_{n=0}^{\nu_{\texttt{reactant}}(p,r)} \left(\frac{[p]}{K_m^{p,r}} \right)^{-n} \right)^{-1} \tag{4}$$

4. $r \in R_{\texttt{reversible}} \land \exists\, e\ (e, r) \in E_R^{\texttt{enzyme}}$, let Equation 5 be the general form of a modular rate law that describes the kinetics of the reaction

$$v_r = [e] \cdot f^r_{\texttt{regulation}} \cdot \frac{N}{D} \tag{5}$$

where

$$N = k^r_{\text{cat},+} \prod_{\substack{(p,r)\\ \in \\ E_R^{\texttt{reactant}}}} \left(\frac{[p]}{K_{m,p}} \right)^{\nu_{\texttt{reactant}}(p,r)} - k^r_{\text{cat},-} \prod_{\substack{(p,r)\\ \in \\ E_R^{\texttt{product}}}} \left(\frac{[p]}{K_{m,p}} \right)^{\nu_{\texttt{product}}(p,r)} \tag{6}$$

$$D = \prod_{\substack{(s,r)\\ \in \\ E_R^{\texttt{reactant}}}} \left(\sum_{n=0}^{\nu_{\texttt{reactant}}(s,r)} \left(\frac{[s]}{K_{m,s}} \right)^{n} \right) + \prod_{\substack{(s,r)\\ \in \\ E_R^{\texttt{product}}}} \left(\sum_{n=0}^{\nu_{\texttt{product}}(s,r)} \left(\frac{[s]}{K_{m,s}} \right)^{n} \right) - 1 \tag{7}$$

In the "Systems Biology" book (Klipp, Edda, Kowald, Axel, pages 49-50) it says that the the denominator D can be one of the following:

1. Power-law modular rate law: $D = 1$
2. Common modular rate law (Equation 7)
3. Simultaneous binding modular rate law

$$D = \prod_{\substack{(p,r)\\ \in \\ E_R^{\texttt{reactant}}}} \left(1 + \frac{[p]}{K_m^{p,r}} \right)^{\nu_{\texttt{reactant}}(p,r)} \prod_{\substack{(p,r)\\ \in \\ E_R^{\texttt{product}}}} \left(1 + \frac{[p]}{K_m^{p,r}} \right)^{\nu_{\texttt{product}}(p,r)} \tag{8}$$

4. Direct binding modular rate law:

$$D = 1 + \prod_{\substack{(p,r)\\ \in \\ E_R^{\texttt{reactant}}}} \left(\frac{[p]}{K_m^{p,r}}\right)^{\nu_{\texttt{reactant}}(p,r)} + \prod_{\substack{(p,r)\\ \in \\ E_R^{\texttt{product}}}} \left(\frac{[p]}{K_m^{p,r}}\right)^{\nu_{\texttt{product}}(p,r)} \tag{9}$$

5. Force-dependent modular rate law:

$$D = \sqrt{\prod_{\substack{(p,r)\\ \in \\ E_R^{\texttt{reactant}}}} \left(1 + \frac{[p]}{K_m^{p,r}}\right)^{\nu_{\texttt{reactant}}(p,r)} \prod_{\substack{(p,r)\\ \in \\ E_R^{\texttt{product}}}} \left(1 + \frac{[p]}{K_m^{p,r}}\right)^{\nu_{\texttt{product}}(p,r)}} \tag{10}$$

At this point I think I'll work with Equation 7, but allow the tool to be configured in order to specify which rate law to use.

This can already be done in the current version of the code, but the only rate law supported is the Mass Action rule.

The definition below can be definetily improved. Note that some object must be removed too, not only added (e.g. if an entity with multiple compartments is split int two entities, then it must be removed from the graph).

NOTE: would it be better to write an algorithm here instead of doing this cumbersome definition?

Definition 4 ("Restructured" biochemical network)**.** Given a network $G = (\mathrm{X}, \mathrm{R}, \mathrm{C}, \mathrm{E}_\mathrm{R}, \mathrm{E}_\mathrm{C})$ let $G' = (\mathrm{X}', \mathrm{R}', \mathrm{C}, \mathrm{E}_\mathrm{R}, \mathrm{E}_\mathrm{C})$ be the normalized version of G where:

- $f_\mathrm{R} : \mathrm{X} \times \mathrm{T} \to \mathrm{R}$ where
  - $f_\mathrm{R}(x, \texttt{role}) = \{r \mid (x, \texttt{role}) \in \mathrm{E}_\mathrm{R}(r)\}$
- $\mathrm{R}' = \mathrm{R} \cup \mathrm{R}_{\texttt{input}} \cup \mathrm{R}_{\texttt{output}}$ where
  - $\mathrm{R}_{\texttt{input}} = \{r_x \mid f_\mathrm{R}(x, \texttt{product}) = \varnothing \land f_\mathrm{R}(x, \texttt{reactant}) \neq \varnothing\}$
  - $\mathrm{R}_{\texttt{output}} = \{r_x \mid f_\mathrm{R}(x, \texttt{reactant}) = \varnothing \land f_\mathrm{R}(x, \texttt{product}) \neq \varnothing\}$

Definition 5 (Biological model satisfiability problem)**.** Given a scenario $\mathcal{S}$ and a normalized biochemical network G let $\langle X, D, C \rangle$ be the satisfiability problem where:

$$\begin{aligned} X = {} & \{K_A^{p,r} \mid r \in R \land \exists p\ (p,r) \in E_R^{\texttt{activator}})\} \\ & \cup \{K_I^{p,r} \mid r \in R \land \exists p\ (p,r) \in E_R^{\texttt{inhibitor}})\} \\ & \cup \{k_+^r \mid r \in R \land \neg \exists e\ (e,r) \in E_R^{\texttt{enzyme}})\} \\ & \cup \{k_-^r \mid r \in R_{\texttt{reversible}} \land \neg \exists e\ (e,r) \in E_R^{\texttt{enzyme}})\} \\ & \cup \left\{k_{\mathrm{cat},+}^r \mid r \in R \land \exists e\ (e,r) \in E_R^{\texttt{enzyme}})\right\} \\ & \cup \left\{k_{\mathrm{cat},-}^r \mid r \in R_{\texttt{reversible}} \land \exists e\ (e,r) \in E_R^{\texttt{enzyme}})\right\} \\ & \cup \{k_m^{p,r} \mid (p,r) \in E_R^{\texttt{reactant}} \land \exists e\ (e,r) \in E_R^{\texttt{enzyme}})\} \\ & \cup \left\{k_m^{p,r} \mid (p,r) \in E_R^{\texttt{product}} \land \exists e\ (e,r) \in E_R^{\texttt{enzyme}}\right)\} \end{aligned} \tag{11}$$

$$D = \{\text{domains here....}\} \tag{12}$$

$$\forall p \in P \quad 0 \leq [p] \leq 1 \tag{13}$$

$$\begin{aligned}&\forall k_1^r, k_2^r \in X, p \in P \\ &\quad (p, r_1, i) \in E \wedge (s, r_2) \in (E_{m^+} \cup E_{m^-}) \wedge r_1 \neq r_2 \rightarrow k_{r_1} < k_{r_2}\end{aligned} \tag{14}$$

$$\forall p \in P_{\text{avg}} \quad [p](\varphi \cdot T) - [p](T) = 0 \tag{15}$$

Definition 6 (Optimization problem)**.** Basically take the satisfiability problem above and turn it in a “single objective optimization problem”

2 Reactome quirks

2.1 Compartments

2.1.1 Physical entities with multiple compartments

Because the functions of biologic molecules critically depend on their subcellular locations, **chemically identical entities located in different compartments are represented as distinct physical entities**. The problem is that many physical entities are related to multiple compartments, even though they are already an instance of an entity related to a compartment.

```
MATCH
    (physicalEntity:PhysicalEntity),
    (physicalEntity)-[:compartment]->(compartment1:Compartment),
    (physicalEntity)-[:compartment]->(compartment2:Compartment)
WHERE compartment1.dbId < compartment2.dbId
OPTIONAL MATCH (compartment1)-[relation]-(compartment2)
WITH
    DISTINCT physicalEntity,
    COLLECT(
        DISTINCT CASE
            WHEN relation IS NOT NULL THEN TYPE(relation)
            ELSE "none"
        END
    ) AS links
RETURN
    apoc.coll.subtract(
        LABELS(physicalEntity),
        [
            "DatabaseObject", "Deletable", "Trackable",
            "PhysicalEntity", "EntitySet"
        ]
    ),
    links,
    COUNT(*)
```

Listing 1: Type of links between compartments of physical entities with multiple compartments.

entity type	compartments link types	#
`CandidateSet`	none	286
`CandidateSet`	*surroundedBy*	29
`CandidateSet`	none, *surroundedBy*	3
`DefinedSet`	none	201
`DefinedSet`	*instanceOf*	1
`DefinedSet`	*surroundedBy*	129
`Polymer`	*surroundedBy*	7

If a `Polymer` involves multiple compartments (or, more generally, a `PhysicalEntity` which is not an `EntitySet`) then:

1. Add new physcal entities, one for each compartment
2. Add reactions with infinite speed that move substance between compartments

```
MATCH (entitySet:EntitySet)
WHERE EXISTS {
    MATCH
        (entitySet)-[:hasMember|hasCandidate]->(physicalEntity:PhysicalEntity)
    WHERE
        NOT EXISTS {
            MATCH
                (entitySet)-[rel1:compartment|includedLocation]-
                (compartment:Compartment)
                -[rel2:compartment|includedLocation]-(physicalEntity)
        }
}
RETURN COUNT(DISTINCT entitySet) // 0
```

Listing 2: There are no instances of `EntitySet` which don't share compartments with their members.

If the relation type includedLocation is not used in Listing 2 then there is an instance of a `Complex` that doesn't share its compartment with the set its placed in.

2.1.2 Physical entities without compartments

Some physical entities do not have compartments. The good news is that these entities are only 25 and are of type `Cell` *(the total number of cells in reactome is 25)*. Moreover, all the reactions that involve cells are of type `CellDevelopmentStep`

```
MATCH (physicalEntity:PhysicalEntity)
WHERE NOT EXISTS {
    MATCH (physicalEntity)-[:compartment]->(:Compartment)
}
OPTIONAL MATCH path = (physicalEntity)--(:ReactionLikeEvent)
RETURN physicalEntity, path
```

Listing 3: Phyiscal entities without any *compartment* and their reactions.

To handle this case it's enough to add a default compartment.

2.1.3 On the compartments of reactions

```
MATCH (reaction:ReactionLikeEvent)
WHERE EXISTS {
    MATCH
        (reaction)--(compartment1:Compartment),
        (reaction)--(compartment2:Compartment)
    WHERE
        compartment1 <> compartment2
}
RETURN COUNT (DISTINCT reaction) // 36322 (out of 93672)
```

Listing 4: Reactions connected to multiple compartments

The fact that reactions are connected to multiple compartments is not a problem! In SBML the compartment is optional for reactions, as the simulators should be able to deduce the compartment of the reaction based on the compartment of its substrate.

2.2 Reactions

2.2.1 Reactions with multiple enzymes

```
MATCH (reaction:ReactionLikeEvent)
WHERE EXISTS {
    MATCH
        (reaction)--(activity1:CatalystActivity),
        (reaction)--(activity2:CatalystActivity)
    WHERE
        activity1 <> activity2
}
RETURN COUNT(DISTINCT reaction) // 17 reactions out of 93672
```

Listing 5: Reactions with multiple `CatalystActivity`

The interesting fact about these reactions is that all of them are of type `BlackBoxEvent`. It might be interesting to add an option to exclude certain types of reactions. In this case a new reaction for each enzyme can be created.

2.2.2 On the reversibility of reactions

```
MATCH (:ReactionLikeEvent)-[relation]-(:ReactionLikeEvent)
RETURN DISTINCT type(relation)
```

Listing 6: Types of links between reactions:
[*precedingEvent*, *inferredTo*, *normalReaction*, *reverseReaction*]

Explanation:

- *precedingEvent*: useful for visualization
- *inferredTo*: reactions inferred from reactions in a different species
- *normalReaction*: it reference to the normal version of a `FailedReaction`
- *reverseReaction*: reference to another reaction which is the reverse

```
MATCH (reaction:ReactionLikeEvent)-[:reverseReaction]->(:ReactionLikeEvent)
RETURN COUNT(DISTINCT reaction) // 112
```

Listing 7: Number of reactions which have a reference to a reverse reaction

Such reactions are marked as `(Reversible)` in the PathwayBrowser

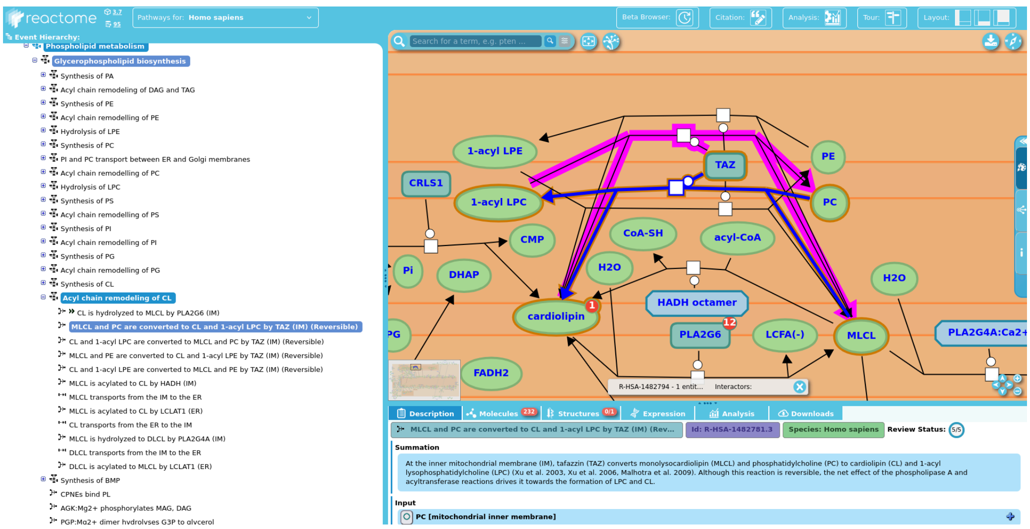

Figure 1: Reversible reaction in the PathwayBrowser: link to view, link to reaction page

```
MATCH (reaction:ReactionLikeEvent)
UNWIND keys(reaction) AS key
RETURN DISTINCT key
```

Listing 8: Attributes of a generic `ReactionLikeEvent`.

The attributes are: `schemaClass`, `speciesName`, `isInDisease`, `releaseDate`, `displayName`, `stIdVersion`, `dbId`, `name`, `isChimeric`, `category`, `stId`, `isInferred`, `oldStId`, `releaseStatus`, `definition`, `systematicName`.

Non of them give information about the “reversibility” of the reaction.

In the official exporter of `Reactome` the reversibility option is set to `false` by default: link to code.

I couldn’t find no other way to determine if a reaction is reversible or not, so only those 112 reaction can be considered reversible.

2.3 Roles physical entities have in reactions

2.3.1 Reactions classification based on the presence of reactants, products, enzymes and regulators

```
MATCH
    (reaction:ReactionLikeEvent)
OPTIONAL MATCH (reaction)-[relation]-(entity)
WHERE
    NOT type(relation) IN [
        "precedingEvent", "negativePrecedingEvent", "normalReaction",
        "inferredTo", "reverseReaction"
    ]
    AND size(apoc.coll.intersection(
        [
            "ControlledVocabulary", "ReviewStatus", "UpdateTracker",
            "InstanceEdit", "Publication", "LiteratureReference",
            "ExternalOntology", "EvidenceType", "URL", "Pathway",
            "Figure", "Anatomy", "Book", "MetaDatabaseObject",
            "DatabaseIdentifier", "RegulationReference",
            "CatalystActivityReference", "EntityFunctionalStatus",
            "DrugActionType", "GO_CellularComponent", "GO_Term",
            "GO_BiologicalProcess", "Species", "Taxon", "Summation",
            "Compartment"
        ],
        labels(entity)
    )) = 0
WITH
    reaction,
    apoc.coll.subtract(
        labels(entity),
        [
            "DatabaseObject", "Deletable", "Trackable",

            "ReactionLikeEvent", "Event",

            "EntitySet", "CandidateSet", "DefinedSet", "SimpleEntity",
            "EntityWithAccessionedSequence", "GenomeEncodedEntity",
            "OtherEntity", "Complex", "Disease",
            "Polymer", "ProteinDrug", "Cell", "Requirement",
            "Drug", "RNADrug", "ChemicalDrug",

            "NegativeRegulation", "PositiveRegulation",
            "NegativeGeneExpressionRegulation",
            "PositiveGeneExpressionRegulation"
        ]
    ) as labels
UNWIND labels AS label
WITH reaction, label ORDER BY label
WITH DISTINCT reaction, COLLECT(DISTINCT label) AS types
RETURN
    apoc.coll.subtract(
        labels(reaction),
        [
            "DatabaseObject", "Deletable", "Trackable",
            "Event", "ReactionLikeEvent"
        ]
    ) AS labels,
    types,
    COUNT(*) AS cnt
ORDER BY labels DESC, size(types)
```

In the table below reaction types are classified by the roles of the reagents in the reaction (where PhysicalEntity groups inputs and outputs). The rows marked in red indicate the reactions that have regulators (inhibitors and activators) but no enzyme.

Reaction	PhysicalEntity	33477
Reaction	CatalystActivity, PhysicalEntity	43889
Reaction	PhysicalEntity, Regulation	2064
Reaction	CatalystActivity, PhysicalEntity, Regulation	3244
Polymerisation	PhysicalEntity	178
Polymerisation	CatalystActivity, PhysicalEntity	36
Polymerisation	PhysicalEntity, Regulation	15
Polymerisation	CatalystActivity, PhysicalEntity, Regulation	10
FailedReaction	PhysicalEntity	171
FailedReaction	CatalystActivity, PhysicalEntity	271
FailedReaction	PhysicalEntity, Regulation	5
FailedReaction	CatalystActivity, PhysicalEntity, Regulation	2
Depolymerisation	PhysicalEntity	3
Depolymerisation	CatalystActivity, PhysicalEntity	25
CellDevelopmentStep	PhysicalEntity	3
CellDevelopmentStep	PhysicalEntity, Regulation	19
BlackBoxEvent	PhysicalEntity	5135
BlackBoxEvent	CatalystActivity, PhysicalEntity	2746
BlackBoxEvent	PhysicalEntity, Regulation	2144
BlackBoxEvent	CatalystActivity, PhysicalEntity, Regulation	235